STATISTICS & MACHINE LEARNING · LESSON 5 OF 28

Hypothesis Testing II: Chi-Square, ANOVA, Non-Parametric Tests and Correlation Tests

Test relationships between categories with chi-square tests, compare three or more groups with one-way ANOVA and Tukey post-hoc tests, use non-parametric alternatives (Mann-Whitney, Wilcoxon, Kruskal-Wallis) when assumptions fail, test correlations, and control false discoveries when running many tests.

ML classes 1 and 3 · Python class 15 | ✦ SciPy 1.1x · statsmodels 0.1x | Intermediate | ⏱ 110 min

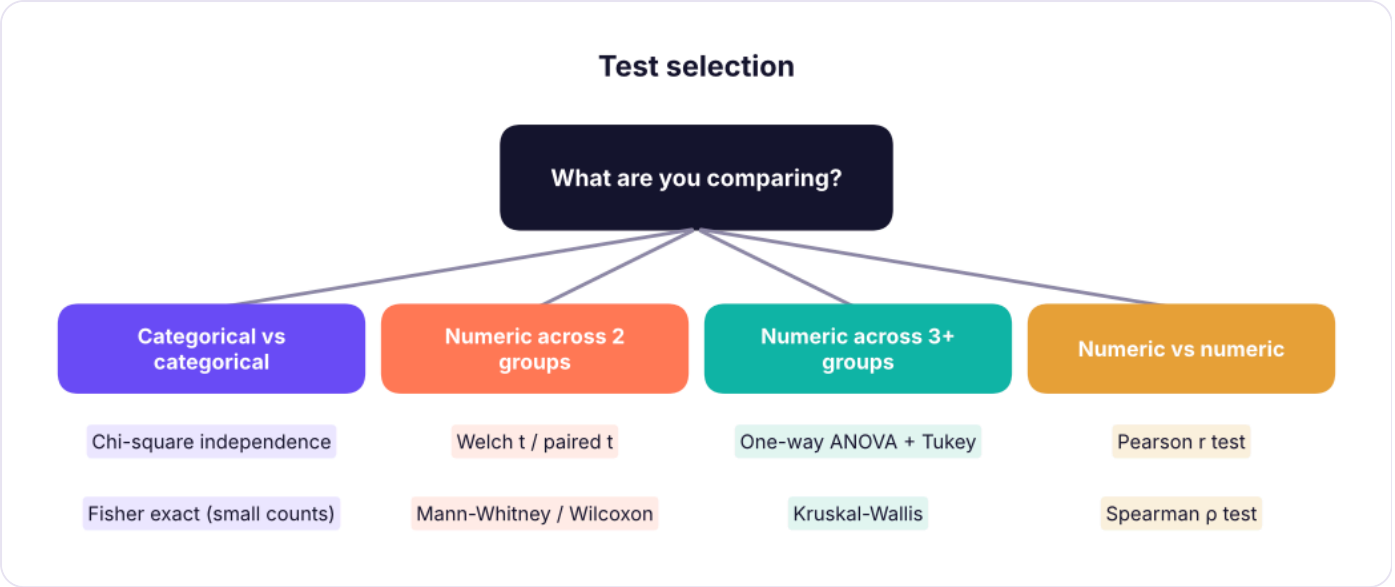
By the end of this lesson you can

- Run chi-square tests of independence and goodness of fit and read expected counts
- Compare several group means with one-way ANOVA and locate differences with Tukey HSD
- Choose parametric vs non-parametric tests and run Mann-Whitney, Wilcoxon and Kruskal-Wallis
- Test correlations with Pearson and Spearman
- Adjust for multiple comparisons (Bonferroni, Benjamini-Hochberg)

Datasets used

`channel_survey.csv` `diet_study.csv` `hr_employees.csv`

Which test? A decision map



Choosing a test from the type of variables and the number of groups.

Parametric (assumes ~normal)	Non-parametric (ranks; fewer assumptions)
One-sample / paired t	Wilcoxon signed-rank
Independent t (Welch)	Mann-Whitney U
One-way ANOVA	Kruskal-Wallis
Pearson correlation	Spearman correlation

Parametric tests are more powerful when their assumptions hold; non-parametric tests are safer for small, skewed samples, ordinal data and outliers.

Chi-square test of independence

> **Case.** A retailer surveyed 600 customers: is the **preferred shopping channel** (App, Store, Website) related to **age group**?

Python

```python
import pandas as pd, numpy as np
from scipy import stats
c = pd.read_csv("channel_survey.csv")
obs = pd.crosstab(c.age_group, c.preferred_channel)
print(obs, "\n")
print((pd.crosstab(c.age_group, c.preferred_channel, normalize="index") * 100).round(1))
res = stats.chi2_contingency(obs)
print(f"\nx² = {res.statistic:.2f}, df = {res.dof}, p = {res.pvalue:.2e}")
print("expected counts if independent:\n", pd.DataFrame(res.expected_freq, index=obs.index,
columns=obs.columns).round(1))
n = obs.values.sum(); cramers_v = np.sqrt(res.statistic / (n * (min(obs.shape) - 1)))
print(f"Cramér's V = {cramers_v:.2f} (effect size: 0.1 small, 0.3 medium, 0.5 large)")
```

OUTPUT

```
preferred_channel  App  Store  Website
age_group
18-25              106     47       65
26-40               72     70       90
41+                 31     56       63

preferred_channel   App  Store  Website
age_group
18-25              48.6   21.6     29.8
26-40              31.0   30.2     38.8
41+                20.7   37.3     42.0

x² = 33.67, df = 4, p = 8.71e-07
expected counts if independent:
 preferred_channel   App  Store  Website
age_group
18-25              75.9   62.9     79.2
26-40              80.8   66.9     84.3
41+                52.2   43.2     54.5
Cramér's V = 0.17 (effect size: 0.1 small, 0.3 medium, 0.5 large)
```

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

The test compares **observed** counts with the counts **expected** if age and channel were independent. p < 0.001, so channel preference depends on age: 18–25-year-olds favour the app (49%), customers over 41 favour stores and the website. Cramér's V ≈ 0.17 shows a modest association.



Standardised residuals show which cells drive the result: positive = more than expected.

Assumptions: independent observations and **expected** counts of at least 5 in (almost) every cell; otherwise combine categories or use Fisher's exact test (`stats.fisher_exact` for 2 × 2).

Goodness of fit

Does an observed distribution match an expected one?

Python

```
shares_expected = np.array([0.35, 0.30, 0.35])                # last year's channel mix
observed = obs.sum().reindex(["App", "Store", "Website"]).to_numpy()
gof = stats.chisquare(observed, f_exp=shares_expected * observed.sum())
print(observed, (shares_expected * observed.sum()).round(0), f"x² = {gof.statistic:.2f}, p = {gof.pvalue:.3f}")
```

OUTPUT

```
[209 173 218] [210. 180. 210.] x² = 0.58, p = 0.748
```

One-way ANOVA

> **Case.** 120 volunteers were randomly assigned to Keto, Low-fat or Mediterranean diets for 8 weeks. Do average weight losses differ?

Running three t-tests inflates the false-positive rate ($1 - 0.95^3 \approx 14\%$). **ANOVA** tests all means at once: H_0: $\mu_1 = \mu_2 = \mu_3$.

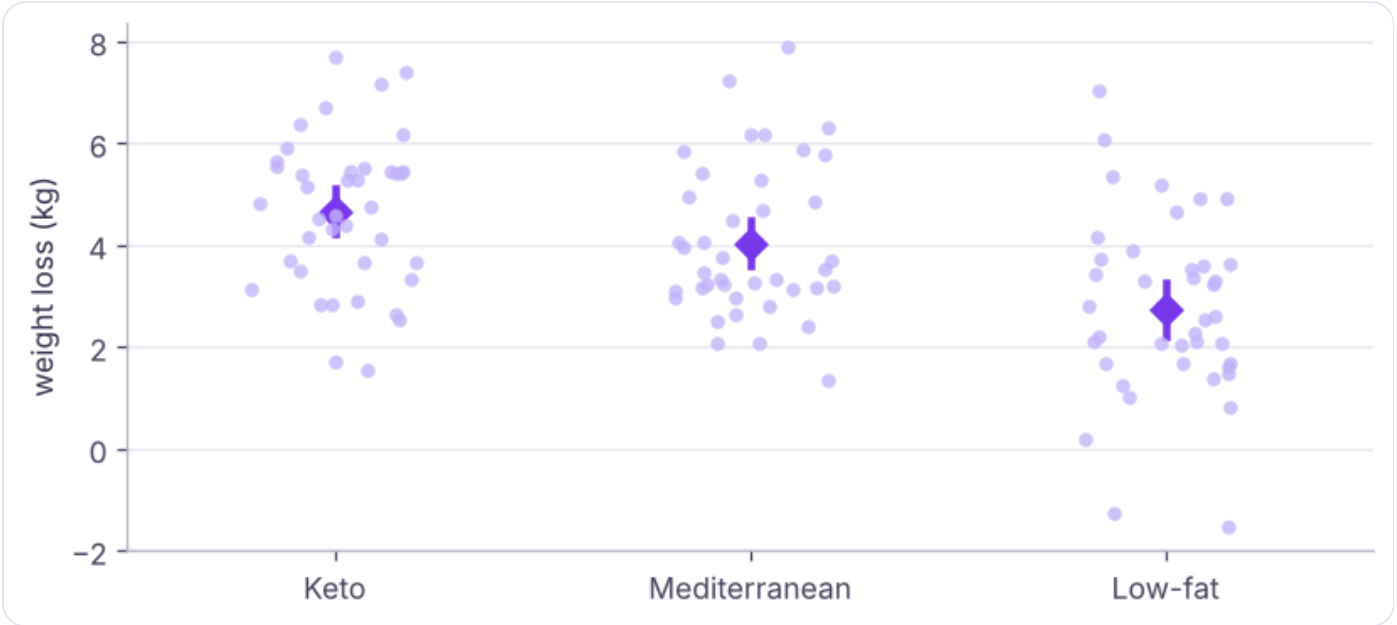
Python

```
d = pd.read_csv("diet_study.csv")
print(d.groupby("diet").weight_loss_kg.agg(["count", "mean", "std"]).round(2))
groups = [g.weight_loss_kg for _, g in d.groupby("diet")]
print("Levene equal variances p =", round(stats.levene(*groups).pvalue, 3), "| Shapiro p
per group:", [round(float(stats.shapiro(g).pvalue), 3) for g in groups])
f = stats.f_oneway(*groups)
print(f"F = {f.statistic:.2f}, p = {f.pvalue:.2e}")
grand = d.weight_loss_kg.mean()
ss_between = sum(len(g) * (g.mean() - grand) ** 2 for g in groups); ss_total =
((d.weight_loss_kg - grand) ** 2).sum()
print(f"n² (share of variance explained by diet) = {ss_between / ss_total:.2f}")
```

OUTPUT

```
               count  mean   std
diet
Keto              40  4.65  1.49
Low-fat           40  2.75  1.78
Mediterranean     40  4.03  1.50
Levene equal variances p = 0.615 | Shapiro p per group: [0.655, 0.698, 0.036]
F = 14.74, p = 1.96e-06
n² (share of variance explained by diet) = 0.20
```

ANOVA splits total variation into **between-group** and **within-group** parts; F = (between variance) ÷ (within variance). A large F means group means differ more than random noise would suggest.



Weight loss by diet with group means: Keto and Mediterranean beat Low-fat.

Post-hoc: which groups differ?

Python

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd
print(pairwise_tukeyhsd(d.weight_loss_kg, d.diet, alpha=0.05))
```

OUTPUT

```
    Multiple Comparison of Means - Tukey HSD, FWER=0.05
=============================================================
 group1     group2    meandiff p-adj   lower   upper  reject
-------------------------------------------------------------
   Keto       Low-fat  -1.9012    0.0 -2.7496 -1.0529   True
   Keto Mediterranean  -0.6152 0.2015 -1.4636  0.2331  False
Low-fat Mediterranean    1.286 0.0014  0.4376  2.1344   True
-------------------------------------------------------------
```

Tukey's HSD compares every pair while keeping the overall error at 5%. Keto and Mediterranean both beat Low-fat; Keto vs Mediterranean is not significant.

Two-way ANOVA (two factors and their interaction) uses `statsmodels.formula.api.ols("y ~ C(a) * C(b)")` with `anova_lm`.

Non-parametric alternatives

Python

```
kw = stats.kruskal(*groups); print(f"Kruskal-Wallis (diets): H = {kw.statistic:.2f}, p =
{kw.pvalue:.2e}")
hr = pd.read_csv("hr_employees.csv")
left, stay = hr[hr.attrition == "Yes"].monthly_income, hr[hr.attrition ==
"No"].monthly_income
print(f"income medians: left {left.median():,.0f} vs stayed {stay.median():,.0f} | Mann-
Whitney p = {stats.mannwhitneyu(left, stay).pvalue:.2e}")
sat_ot = hr[hr.overtime == "Yes"].job_satisfaction; sat_no = hr[hr.overtime ==
"No"].job_satisfaction
print(f"job satisfaction (ordinal 1-4), overtime vs not: Mann-Whitney p =
{stats.mannwhitneyu(sat_ot, sat_no).pvalue:.3f}")
rng = np.random.default_rng(1); before = rng.lognormal(3, .5, 20); after = before *
rng.uniform(.75, 1.05, 20)
print("Wilcoxon signed-rank (paired, skewed):", round(stats.wilcoxon(before, after).pvalue,
4))
```

OUTPUT

```
Kruskal-Wallis (diets): H = 23.29, p = 8.78e-06
income medians: left 67,917 vs stayed 77,750 | Mann-Whitney p = 3.81e-06
job satisfaction (ordinal 1–4), overtime vs not: Mann-Whitney p = 0.191
Wilcoxon signed-rank (paired, skewed): 0.001
```

Mann-Whitney compares whether values in one group tend to be larger than in the other (often described as comparing medians). It suits skewed income data and ordinal ratings.

Testing correlations

Python

```python
for a, b in [("age", "monthly_income"), ("distance_from_home_km", "job_satisfaction"),
("years_at_company", "monthly_income")]:
    pr, sp = stats.pearsonr(hr[a], hr[b]), stats.spearmanr(hr[a], hr[b])
    print(f"{a:22s} vs {b:16s} Pearson r {pr.statistic:+.3f} (p {pr.pvalue:.3g}) | Spearman
ρ {sp.statistic:+.3f} (p {sp.pvalue:.3g})")
print("Department vs attrition: x² p =",
round(stats.chi2_contingency(pd.crosstab(hr.department, hr.attrition)).pvalue, 3))
```

OUTPUT

```
age                   vs monthly_income   Pearson r +0.135 (p 1.52e-07) | Spearman ρ
+0.108 (p 2.95e-05)
distance_from_home_km  vs job_satisfaction Pearson r +0.008 (p 0.764) | Spearman ρ
+0.007 (p 0.776)
years_at_company      vs monthly_income   Pearson r +0.679 (p 1.63e-203) | Spearman
ρ +0.577 (p 7.98e-134)
Department vs attrition: x² p = 0.196
```

With 1,500 rows, even r = 0.1 can be "significant": look at the size of r, not only the p-value.

Many tests: the multiple-comparison problem

Test 100 segments where nothing really changed, at α = 0.05, and about five will look "significant" by chance.

Python

```python
from statsmodels.stats.multitest import multipletests
rng = np.random.default_rng(42)
null_p = [stats.ttest_ind(rng.normal(0, 1, 50), rng.normal(0, 1, 50)).pvalue for _ in
range(100)]       # 100 segments with NO real effect
real_p = [stats.ttest_ind(rng.normal(0, 1, 50), rng.normal(0.8, 1, 50)).pvalue for _ in
range(5)]      # 5 segments with a real effect
pvals = np.array(null_p + real_p); truth = np.array([False] * 100 + [True] * 5)
for name, sig in [("raw p < 0.05", pvals < 0.05), ("Bonferroni", multipletests(pvals, 0.05,
"bonferroni")[0]), ("Benjamini-Hochberg", multipletests(pvals, 0.05, "fdr_bh")[0])]:
    print(f"{name:20s} significant {sig.sum():3d} | true effects found {(sig &
truth).sum()} of 5 | false positives {(sig & ~truth).sum()}")
```

OUTPUT

```
raw p < 0.05         significant  12 | true effects found 5 of 5 | false positives 7
Bonferroni           significant   3 | true effects found 3 of 5 | false positives 0
Benjamini-Hochberg   significant   6 | true effects found 5 of 5 | false positives 1
```

Bonferroni (α ÷ number of tests) controls the chance of any false positive but is strict; **Benjamini-Hochberg** controls the false discovery rate and keeps more power. Use them for dashboards that test many segments, A/B tests with many metrics, and feature screening.

Summary

KEY TAKEAWAYS

- Chi-square tests whether two categorical variables are independent (expected counts ≥ 5); report Cramér's V and residuals.
- ANOVA compares 3+ means in one test (F = between/within variance); follow with Tukey HSD; report η^2.
- Non-parametric tests (Mann-Whitney, Wilcoxon, Kruskal-Wallis, Spearman) use ranks for skewed, ordinal or small data.
- Test correlations with Pearson (linear) or Spearman (monotonic); large n makes tiny r significant.
- Many tests → adjust p-values (Bonferroni, Benjamini-Hochberg).

Quick quiz

6 questions · tap an answer to check it

1. Which test checks whether preferred channel depends on age group?

A ANOVA

B Chi-square test of independence

C Paired t-test

D Pearson correlation

Answer: B. Both variables are categorical.

2. Why use ANOVA instead of several t-tests for 3 groups?

A Multiple t-tests inflate the chance of a false positive

B t-tests cannot handle numbers

C ANOVA always gives smaller p-values

D It is required by law

Answer: A. ANOVA tests all means together at α.

3. ANOVA gives p < 0.001 for three diets. What next?

A Stop: all diets differ

B Run a post-hoc test (Tukey HSD) to see which pairs differ

C Run a chi-square test

D Delete outliers

Answer: B. ANOVA only says at least one mean differs.

4. Which test compares two independent groups of skewed incomes?

A Paired t-test

B Mann-Whitney U

C Chi-square goodness of fit

D One-sample t-test

Answer: B. A rank-based alternative to the t-test.

5. What does the Benjamini-Hochberg method control?

A Type II errors

B The false discovery rate among significant results

C The sample size

D Normality

Answer: B. It is less strict than Bonferroni.

6. What is a chi-square assumption?

A Expected counts of at least about 5 per cell

B Normal data

C Equal variances

D Paired observations

Answer: A. Use Fisher's exact test for small tables.

EXERCISE

Practice (hr_employees.csv)

(1) Is overtime related to department? (chi-square + Cramér's V). (2) Does mean monthly income differ across departments (ANOVA + Tukey)? (3) Repeat (2) with Kruskal-Wallis. (4) Is education level related to attrition?

Show solution (worked answer)

Python

```
t = pd.crosstab(hr.department, hr.overtime); r = stats.chi2_contingency(t);
print(r.pvalue, np.sqrt(r.statistic / (t.values.sum() * (min(t.shape) - 1))))
gs = [g.monthly_income for _, g in hr.groupby("department")];
print(stats.f_oneway(*gs))
print(pairwise_tukeyhsd(hr.monthly_income, hr.department))
print(stats.kruskal(*gs))
print(stats.chi2_contingency(pd.crosstab(hr.education, hr.attrition)).pvalue)
```

INTERVIEW CORNER

Questions you may be asked

Q: When would you use a non-parametric test? Small samples that are clearly non-normal, ordinal data (ratings), or heavy outliers; for large samples the CLT usually makes parametric tests fine.

Q: What does a chi-square test tell you and what does it not? Whether two categorical variables are associated; not the direction, strength (use Cramér's V) or causation.

Q: Explain ANOVA's F-statistic. The ratio of variation between group means to variation within groups; large values mean group differences exceed what noise explains.

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